

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Summer Examination – 2023**

**Course: B. Tech.**

**Branch : E&TC**

**Semester : VI**

**Subject Code & Name: BTETPE603A- Microprocessors and Microcontrollers**

**Max Marks: 60**

**Date: 17/07/2023**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                | (Level/CO) | Marks     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q.1 Solve Any Two of the following.</b>                                                                                                                     |            | <b>12</b> |
| <b>A)</b> Provide a detailed explanation of the flag register of an 8085 microprocessor.                                                                       | <b>L6</b>  | <b>6</b>  |
| <b>B)</b> State addressing modes of following instructions of 8085 microprocessor.<br>up: LXI H, 1000H<br>MOV A, M<br>MOV B, A<br>ADD B<br>STA 1001H<br>JMP up | <b>L3</b>  | <b>6</b>  |
| <b>C)</b> State any six differences between CISC and RISC architecture.                                                                                        | <b>L3</b>  | <b>6</b>  |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                     |            | <b>12</b> |
| <b>A)</b> Explain following instructions with suitable examples of each.<br>(i) ANI 55H (ii) CMP B (iii) SBI 55H                                               | <b>L6</b>  | <b>6</b>  |
| <b>B)</b> Assume that five numbers are stored from 1000H to 1004H. Write an assembly language program to clear D3 bit of each number.                          | <b>L4</b>  | <b>6</b>  |
| <b>C)</b> Draw timing diagram for following instructions of 8085.<br>(i) MOV A, B (ii) MVI A, 55H                                                              | <b>L5</b>  | <b>6</b>  |

|                                                                                                                                   |           |           |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                       |           | <b>12</b> |
| <b>A)</b> Draw and explain Pin diagram of an 8051 microcontroller.                                                                | <b>L6</b> | <b>6</b>  |
| <b>B)</b> Explain any three addressing modes of 8051 with suitable examples of each.                                              | <b>L6</b> | <b>6</b>  |
| <b>C)</b> Draw structure of a TMOD register and explain function of each bit of it.                                               | <b>L5</b> | <b>6</b>  |
| <br><b>Q.4 Solve Any Two of the following.</b>                                                                                    |           | <b>12</b> |
| <b>A)</b> Write an 8051 C program to toggle all bits of P2 continuously every 500 ms.<br>Use Timer 1, mode 1 to create the delay. | <b>L4</b> | <b>6</b>  |
| <b>B)</b> Explain any six pins of RS232 DB-9 connector.                                                                           | <b>L1</b> | <b>6</b>  |
| <b>C)</b> Indicate which mode and which timer is selected for each of the following.                                              | <b>L3</b> | <b>6</b>  |
| i.   MOV TMOD, #01H                                                                                                               |           |           |
| ii.   MOVTMOD, #20H                                                                                                               |           |           |
| iii.   MOV TMOD, #12H                                                                                                             |           |           |
| <br><b>Q. 5 Solve Any Two of the following.</b>                                                                                   |           | <b>12</b> |
| <b>A)</b> Write an 8051 C program to send letters ‘A’, ‘B’, and ‘C’ to the LCD using the busy flag method.                        | <b>L1</b> | <b>6</b>  |
| <b>B)</b> Explain BSR and Input-Output modes of 8255 with neat diagrams.                                                          | <b>L5</b> | <b>6</b>  |
| <b>C)</b> Explain internal RAM memory structure of an 8051 microcontroller.                                                       | <b>L3</b> | <b>6</b>  |

**\*\*\* End \*\*\***